Exercise J-M

1. $\tan^{-1}\left(\frac{1+\sqrt{3}}{3+\sqrt{3}}\right)+\sec^{-1}\left(\sqrt{\frac{8+4\sqrt{3}}{6+3\sqrt{3}}}\right)$ is equal to : **[JEE(Main)Jan.2023]**

$\tan^{-1}\left(\frac{1+\sqrt{3}}{3+\sqrt{3}}\right)+\sec^{-1}\left(\sqrt{\frac{8+4\sqrt{3}}{6+3\sqrt{3}}}\right)$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{3}$

Ans. D

Sol. $E=\tan^{-1}\left(\frac{1+\sqrt{3}}{\sqrt{3}(\sqrt{3}+1)}\right)+\sec^{-1}\left(\frac{16+8\sqrt{3}}{12+6\sqrt{3}}\right)^{1/2}$

$\because \frac{16+8\sqrt{3}}{12+6\sqrt{3}}=\frac{4(\sqrt{3}+1)^2}{(3+\sqrt{3})^2}$

$\therefore \left(\frac{16+8\sqrt{3}}{12+6\sqrt{3}}\right)^{1/2}=\frac{2(\sqrt{3}+1)}{3+\sqrt{3}}=\frac{2}{\sqrt{3}}$

$E=\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)+\sec^{-1}\left(\frac{2}{\sqrt{3}}\right)=\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)+\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$

$E=\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)+\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$

$=\frac{\pi}{6}+\frac{\pi}{6}=\frac{\pi}{3}$

2. If $\sin^{-1}\frac{\alpha}{17}+\cos^{-1}\frac{4}{5}-\tan^{-1}\frac{77}{36}$, $0<\alpha<13$ then $\sin^{-1}(\sin\alpha)+\cos^{-1}(\cos\alpha)$ is equal to : **[JEE(Main)Jan.2023]**

यदि $\sin^{-1}\frac{\alpha}{17}+\cos^{-1}\frac{4}{5}-\tan^{-1}\frac{77}{36}$, $0<\alpha<13$ है, तो $\sin^{-1}(\sin\alpha)+\cos^{-1}(\cos\alpha)$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 16 (B) 0 (C) π (D) $16-5\pi$

Ans. C

Sol. $\cos^{-1}\frac{4}{5}=\tan^{-1}\frac{3}{4}$

$$\therefore \sin^{-1}\frac{\alpha}{17} = \tan^{-1}\frac{77}{36} - \tan^{-1}\frac{3}{4} = \tan^{-1}\left(\frac{\frac{77}{36}-\frac{3}{4}}{1+\frac{77}{36}\cdot\frac{3}{4}}\right)$$

$$\sin^{-1}\frac{\alpha}{17} = \tan^{-1}\frac{8}{15} = \sin^{-1}\frac{8}{17}$$

$$\Rightarrow \frac{\alpha}{17} = \frac{8}{17} \Rightarrow \alpha = 8$$

$\therefore \sin^{-1}(\sin 8) + \cos^{-1}(\cos 8)$
$= 3\pi - 8 + 8 - 2\pi$
$= \pi$

3. Let $a_1 = 1, a_2, a_3, a_4, \ldots\ldots$ be consecutive natural numbers.

Then $\tan^{-1}\left(\frac{1}{1+a_1a_2}\right) + \tan^{-1}\left(\frac{1}{1+a_2a_3}\right) + \ldots\ldots + \tan^{-1}\left(\frac{1}{1+a_{2021}a_{2022}}\right)$ is equal to :

[JEE(Main)Jan.2023]

माना $a_1 = 1, a_2, a_3, a_4, \ldots\ldots$ क्रमागत धन पूर्णांक है। तब

$\tan^{-1}\left(\frac{1}{1+a_1a_2}\right) + \tan^{-1}\left(\frac{1}{1+a_2a_3}\right) + \ldots\ldots + \tan^{-1}\left(\frac{1}{1+a_{2021}a_{2022}}\right)$ बराबर है :

[JEE(Main)Jan.2023]

(A) $\tan^{-1}(2022) - \frac{\pi}{4}$ (B) $\frac{\pi}{4} - \cot^{-1}(2022)$ (C) $\frac{\pi}{4} - \tan^{-1}(2022)$ (D) $\cot^{-1}(2022) - \frac{\pi}{4}$

Ans. A, B

Sol. $a_2 - a_1 = a_3 - a_2 = \ldots\ldots = a_{2022} - a_{2021} = 1.$

$$\therefore \tan^{-1}\left(\frac{a_2-a_1}{1+a_1a_2}\right) + \tan^{-1}\left(\frac{a_3-a_2}{1+a_2a_3}\right) + \ldots\ldots + \tan^{-1}\left(\frac{a_{2022}-a_{2021}}{1+a_{2021}a_{2022}}\right)$$

$$\left[\left(\tan^{-1}a_2\right) - \tan^{-1}a_1\right] + \left[\tan^{-1}a_3 - \tan^{-1}a_2\right] + \ldots\ldots + \left[\tan^{-1}a_{2022} - \tan^{-1}a_{2021}\right]$$

$= \tan^{-1}a_{2022} - \tan^{-1}a_1$

$= \tan^{-1}(2022) - \tan^{-1}1 = \tan^{-1}2022 - \frac{\pi}{4}$ **(Option 3)**

$$= \left(\frac{\pi}{2} - \cot^{-1}(2022)\right) - \frac{\pi}{4}$$

$= \frac{\pi}{4} - \cot^{-1}(2022)$ **(Option 1)**

4. Let $(a,b) \subset (0,2\pi)$ be the largest interval for which $\sin^{-1}(\sin\theta) - \cos^{-1}(\sin\theta) > 0$, $\theta \in (0,2\pi)$, holds. If

$\alpha x^2+\beta x+\sin^{-1}\left(x^2-6x+10\right)+\cos^{-1}\left(x^2-6x+10\right)=0$ and $\alpha-\beta=b-a$, then α is equal to :

[JEE(Main)Jan.2023]

माना $(a,b)\subset(0,2\pi)$ सबसे बड़ा अंतराल है, जिसके लिए $\sin^{-1}(\sin\theta)-\cos^{-1}(\sin\theta)>0$, $\theta\in(0,2\pi)$ है। यदि $\alpha x^2+\beta x+\sin^{-1}\left(x^2-6x+10\right)+\cos^{-1}\left(x^2-6x+10\right)=0$ तथा $\alpha-\beta=b-a$ है, तो α बराबर है :

[JEE(Main)Jan.2023]

(A) $\frac{\pi}{48}$ (B) $\frac{\pi}{16}$ (C) $\frac{\pi}{8}$ (D) $\frac{\pi}{12}$

Ans. D

Sol. $\sin^{-1}\sin\theta-\left(\frac{\pi}{2}-\sin^{-1}\sin\theta\right)>0$

$\Rightarrow \sin^{-1}\sin\theta>\frac{\pi}{4}$

$\Rightarrow \sin\theta>\frac{1}{\sqrt{2}}$

So, $\theta\in\left(\frac{\pi}{4},\frac{3\pi}{4}\right)$

$\theta\in\left(\frac{\pi}{4},\frac{3\pi}{4}\right)=(a,b)$

$\Rightarrow b-a=\frac{\pi}{2}=\alpha-\beta$

$\Rightarrow \beta=\alpha-\frac{\pi}{2}$

$\Rightarrow \alpha x^2+\beta x+\sin^{-1}\left[(x-3)^2+1\right]+\cos^{-1}\left[(x-3)^2+1\right]=0$

$x=3, 9\alpha+3\beta+\frac{\pi}{2}+0=0$

$\Rightarrow 9\alpha+3\left(\alpha-\frac{\pi}{2}\right)+\frac{\pi}{2}=0$

$\Rightarrow 12\alpha-\pi=0$

$\alpha=\frac{\pi}{12}$

5. Let S be the set of all solutions of the equation $\cos^{-1}(2x)-2\cos^{-1}\left(\sqrt{1-x^2}\right)=\pi$, $x\in\left[-\frac{1}{2},\frac{1}{2}\right]$. Then $\sum_{x\in S}2\sin^{-1}\left(x^2-1\right)$ is equal to : **[JEE(Main)Jan.2023]**

माना समीकरण $\cos^{-1}(2x)-2\cos^{-1}\left(\sqrt{1-x^2}\right)=\pi,\ x\in\left[-\frac{1}{2},\frac{1}{2}\right]$ के सभी हलों का समुच्चय S है। तब $\sum_{x\in S}2\sin^{-1}\left(x^2-1\right)$ का मान है : **[JEE(Main)Jan.2023]**

(A) 0 (B) $\pi-\sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$ (C) $\frac{-2\pi}{3}$ (D) $\pi-2\sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$

Ans. C

Sol. $\cos^{-1}(2x)=\pi+2\cos^{-1}\left(\sqrt{1-x^2}\right)$

LHS = $[0, \pi]$

For equation to be meaningful

$\cos^{-1}2x=\pi$ and $\cos^{-1}\left(\sqrt{1-x^2}\right)=0$

$x=\frac{-1}{2}$ and $x=0$

which is not possible

$\therefore\ x\in\phi$

Now $\Sigma(x)=0$

$\therefore$ Sum over empty set is always 0

6. Let $S=\left\{x\in R:0<x<1 \text{ and } 2\tan^{-1}\left(\frac{1-x}{1+x}\right)=\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)\right\}$. If n(S) denotes the number of elements in S then: **[JEE(Main)Jan.2023]**

(A) n(S) = 2 and only one element in S is less than $\frac{1}{2}$.

(B) n(S) = 0

(C) n(S) = 1 and the element in S is less than $\frac{1}{2}$.

(D) n(S) = 1 and the elements in S is more than $\frac{1}{2}$.

माना $S=\left\{x\in R:0<x<1 \text{ and } 2\tan^{-1}\left(\frac{1-x}{1+x}\right)=\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)\right\}$ है। यदि n(S), S में अवयवों की संख्या को दर्शाता है, तो: **[JEE(Main)Jan.2023]**

(A) n(S) = 2 है तथा S का एक अवयव $\frac{1}{2}$ से कम है

(B) n(S) = 0

(C) n(S) = 1 है तथा S का अवयव $\frac{1}{2}$ से कम है

(D) n(S) = 1 है तथा S का एक अवयव $\frac{1}{2}$ से अधिक है

Ans. C

Sol. $0 < x < 1$

$$2\tan^{-1}\left(\frac{1-x}{1+x}\right)=\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$$

$$\tan^{-1}x=\theta\in\left(0,\frac{\pi}{4}\right)\ \therefore x=\tan\theta$$

$$2\tan^{-1}\left(\tan\left(\frac{\pi}{4}-\theta\right)\right)=\cos^{-1}\left(\cos 2\theta\right)$$

$$2\left(\frac{\pi}{4}-\theta\right)=2\theta\therefore 4\theta=\frac{\pi}{2}\therefore\theta=\frac{\pi}{8}$$

$$x=\tan\frac{\pi}{8}\therefore x=\sqrt{2}-1\simeq 0.414$$

7. Let $y = f(x)$ represent a paralbola with focus $\left(-\frac{1}{2},0\right)$ and directrix $y=-\frac{1}{2}$. Then

$$S=\left\{x\in R:\tan^{-1}\left(\sqrt{f(x)}\right)+\sin^{-1}\left(\sqrt{f(x)+1}\right)=\frac{\pi}{2}\right\}:$$ **[JEE(Main)Jan.2023]**

(A) is an infinite set (B) is an empty set

(C) contains exactly two elements (D) contains exactly one element

माना $y = f(x)$ नाभि $\left(-\frac{1}{2},0\right)$ तथा नियता $y=-\frac{1}{2}$ के परवलय को निरूपित करता है। तो

$$S=\left\{x\in R:\tan^{-1}\left(\sqrt{f(x)}\right)+\sin^{-1}\left(\sqrt{f(x)+1}\right)=\frac{\pi}{2}\right\}:$$ **[JEE(Main)Jan.2023]**

(A) एक अपरिमित समुच्चय है (B) एक रिक्त समुच्चय है

(C) में ठीक दो अवयव है (D) में केवल एक अवयव है

Ans. C

Sol. Equation of parabola $\sqrt{\left(x+\frac{1}{2}\right)^2+y^2}=\left|y+\frac{1}{2}\right|$

$$x^2+\frac{1}{4}+x+y^2=y^2+\frac{1}{4}+y$$

$x^2 + x + y = f(x)$

Now $\tan^{-1}\sqrt{f(x)}+\sin^{-1}\sqrt{f(x)+1}=\frac{\pi}{2}$

$\cos^{-1}\frac{1}{\sqrt{1+f(x)}}+\sin^{-1}\sqrt{f(x)+1}=\frac{\pi}{2}$

So $\sqrt{f(x)+1}=\frac{1}{\sqrt{1+f(x)}}$

$f(x)+1=1$

$f(x)=0\Rightarrow x=0,-1$

8. If the sum of all the solution of $\tan^{-1}\left(\frac{2x}{1-x^2}\right)+\cot^{-1}\left(\frac{1-x^2}{2x}\right)=\frac{\pi}{3}, -1<x<1, x\neq 0$, is $\alpha-\frac{4}{\sqrt{3}}$, then α is equal to _____. **[JEE(Main)Jan.2023]**

यदि $\tan^{-1}\left(\frac{2x}{1-x^2}\right)+\cot^{-1}\left(\frac{1-x^2}{2x}\right)=\frac{\pi}{3}, -1<x<1, x\neq 0$, के सभी हलों का योग $\alpha-\frac{4}{\sqrt{3}}$ है, तो α बराबर है। **[JEE(Main)Jan.2023]**

Ans. 2

Sol. **Case I :** $x>0$

$\tan^{-1}\frac{2x}{1-x^2}+\tan^{-1}\frac{2x}{1-x^2}=\frac{\pi}{3}$

$x=2-\sqrt{3}$

Case II : $x<0$

$\tan^{-1}\frac{2x}{1-x^2}+\tan^{-1}\frac{2x}{1-x^2}+\pi=\frac{\pi}{3}$

$x=\frac{-1}{\sqrt{3}}\Rightarrow\alpha=2$

9. Let D be the domain of the function $f(x)=\sin^{-1}\left(\log_{3x}\left(\frac{6+2\log_3 x}{-5x}\right)\right)$. If the range of the function $g:D\to\mathbb{R}$ defined by $g(x)=x-[x]$ ([x] is the greatest integer function), is (α,β), then $\alpha^2+\frac{5}{\beta}$ is equal to :

[JEE(Main)April,2023]

माना फलन $f(x)=\sin^{-1}\left(\log_{3x}\left(\frac{6+2\log_3 x}{-5x}\right)\right)$ का प्राँत D है। यदि g(x) = x – [x], ([x] महत्तम पूर्णांक फलन है), द्वारा परिभाषित फलन $g:D\to\mathbb{R}$ का परिसर (α,β) है, तो $\alpha^2+\frac{5}{\beta}$ बराबर है : **[JEE(Main)April,2023]**

(A) 46 (B) 136 (C) 45 (D) 135

Ans. ()

Sol. $\Rightarrow$ From (2), $x>0$, from (3), $x\neq\frac{1}{3}$

From (4), $6+2\log_3 x<0 \quad (\because x>0)$

$\log_3 x<-3n$

$\Rightarrow x<3^{-3}n$

$\Rightarrow \quad x<\frac{1}{27}$

$\Rightarrow$ From (2), (3), (4), $0<x<\frac{1}{27}$ .....(5)

From (1), $-1\le\log_{3x}\left(\frac{6+2\log_3 x}{-5x}\right)\le 1$

$\because \quad 3x\in\left(0,\frac{1}{9}\right)$

$\Rightarrow \frac{1}{3x}\ge\frac{6+2\log_3 x}{-5x}\ge 3x$

$\Rightarrow -\frac{5}{3}\le 6+2\log_3 x\le -15x^2$

$\Rightarrow -\frac{23}{6}\le\log_3 x\le\frac{-15x^2-6}{2}$

$\Rightarrow 3^{-\frac{23}{6}}\le x\le 3^{\frac{-15x^2-6}{2}}$

$\Rightarrow 3^{\frac{1}{23}}\le x\le\frac{1}{3^{\frac{15x^2+6}{2}}}$

From (5) & (6), $0 < x < \frac{1}{27}, [x] = 0$

$\Rightarrow g(x) = x$

$\Rightarrow$ Range g(x) is domain of f(x)

10. The range of $f(x) = 4\sin^{-1}\left(\frac{x^2}{x^2+1}\right)$ is : **[JEE(Main)April,2023]**

$f(x) = 4\sin^{-1}\left(\frac{x^2}{x^2+1}\right)$ का परिसर है : **[JEE(Main)April,2023]**

(A) $[0, \pi]$ (B) $[0, 2\pi)$ (C) $[0, 2\pi]$ (D) $[0, \pi)$

Ans. B

Sol. $f(x) = 4\sin^{-1}\left(\frac{x^2}{1+x^2}\right)$

$0 \le \frac{x^2}{1+x^2} < 1$

$\Rightarrow 0 \le \sin^{-1}\left(\frac{x^2}{1+x^2}\right) < \frac{\pi}{2}$

$\Rightarrow 0 \le 4\sin^{-1}\left(\frac{x^2}{1+x^2}\right) < 2\pi$

Range : $[0, 2\pi)$

11. If the domain of the function $f(x) = \log_e\left(4x^2+11x+6\right) + \sin^{-1}(4x+3) + \cos^{-1}\left(\frac{10x+6}{3}\right)$ is $(\alpha, \beta]$, then $36|\alpha+\beta|$ is equal to : **[JEE(Main)April,2023]**

यदि $f(x) = \log_e\left(4x^2+11x+6\right) + \sin^{-1}(4x+3) + \cos^{-1}\left(\frac{10x+6}{3}\right)$ का प्रांत $(\alpha, \beta]$ है, तो $36|\alpha+\beta|$ का मान बराबर होगा : **[JEE(Main)April,2023]**

(A) 63 (B) 54 (C) 45 (D) 72

Ans. C

Sol. $4x^2 + 11x + 6 > 0 \Rightarrow (4x+3)(x+2) > 0$

$\Rightarrow x \in (-\infty, -2) \cup \left(\frac{-3}{4}, \infty\right)$ ...(i)

$-1 \le 4x+3 \le 1 \Rightarrow -4 \le 4x \le -2$

$$-1 \le x \le -\frac{1}{2} \qquad \text{...(ii)}$$

$$-1 \le \frac{10x+6}{3} \le 1 \Rightarrow -3 \le 10x+6 \le 3$$

$$\Rightarrow -9 \le 10x \le -3$$

$$\frac{-9}{10} \le x \le \frac{-3}{10} \qquad \text{...(iii)}$$

$$(i),(ii),(iii) \Rightarrow \frac{-3}{4} < x \le \frac{-1}{2}$$

$$\alpha = \frac{-3}{4}, \beta = \frac{-1}{2}$$

$$36|\alpha+\beta| = 36 \times \frac{5}{4} = 45$$

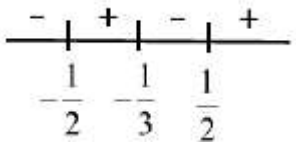
12. If domain of the function $\log_e\left(\frac{6x^2+5x+1}{2x-1}\right)+\cos^{-1}\left(\frac{2x^2-3x+4}{3x-5}\right)$ is $(\alpha, \beta) \cup (\gamma, \delta]$, then $18(\alpha^2+\beta^2+\gamma^2+\delta^2)$ is equal to _____. **[JEE(Main)April,2023]**

यदि फलन $\log_e\left(\frac{6x^2+5x+1}{2x-1}\right)+\cos^{-1}\left(\frac{2x^2-3x+4}{3x-5}\right)$ का प्राँत $(\alpha, \beta) \cup (\gamma, \delta]$ है, तो $18(\alpha^2+\beta^2+\gamma^2+\delta^2)$ बराबर है : **[JEE(Main)April,2023]**

Ans. (20)

Sol. $$\frac{6x^2+5x+1}{2x-1} > 0$$

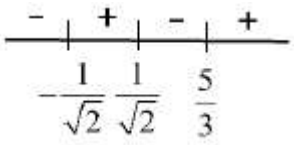
$$\frac{(3x+1)(2x+1)}{2x-1} > 0$$



$$x \in \left(\frac{-1}{2}, \frac{-1}{3}\right) \cup \left(\frac{1}{2}, \infty\right) \qquad \text{....(A)}$$

and $$-1 \le \frac{2x^2-3x+4}{3x-5} \le 1$$

$$\frac{2x^2-1}{3x-5} \ge 0 \quad \text{and} \quad \frac{2x^2-6x+9}{3x-5} \le 0$$



and $3x - 5 < 0$

$x \in \left[\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right] \cup \left(\frac{5}{3}, \infty\right)$ ....(B)

$x < \frac{5}{3}$ ....(C)

$A \cap B \cap C \equiv \left(\frac{-1}{2}, \frac{-1}{3}\right) \cup \left(\frac{1}{2}, \frac{1}{\sqrt{2}}\right]$

So $18\left(\alpha^2 + \beta^2 + \gamma^2 + \delta^2\right) = 18\left(\frac{1}{4} + \frac{1}{9} + \frac{1}{4} + \frac{1}{2}\right)$

$= 18 + 2 = 20$

13. If the domain of the function $f(x) = \sec^{-1}\left(\frac{2x}{5x+3}\right)$ is $[\alpha, \beta) \cup (\gamma, \delta]$, then $|3\alpha + 10(\beta + \gamma) + 21\delta|$ is equal to _____. **[JEE(Main)April,2023]**

यदि फलन $f(x) = \sec^{-1}\left(\frac{2x}{5x+3}\right)$ का प्राँत $[\alpha, \beta) \cup (\gamma, \delta]$ है, तो $|3\alpha + 10(\beta + \gamma) + 21\delta|$ बराबर है : **[JEE(Main)April,2023]**

Ans. (24.00)

Sol. $\frac{2x}{5x+3} \geq 1$ OR ≤ -1

$\frac{-3x-3}{5x+3} \geq 0$ $\frac{7x+3}{5x+3} \leq 0$

$\frac{x+1}{5x+3} \leq 0$ $x \in \left(\frac{-3}{5}, \frac{-3}{7}\right]$

$x \in \left[-1, -\frac{3}{5}\right)$

$3\alpha = -3, 10(\beta + \gamma) = -12, 21\delta = -9$

14. If $S = \left\{x \in \mathbb{R} : \sin^{-1}\left(\frac{x+1}{\sqrt{x^2+2x+2}}\right) - \sin^{-1}\left(\frac{x}{\sqrt{x^2+1}}\right) = \frac{\pi}{4}\right\}$, then

$\sum_{x \in s}\left(\sin\left(\left(x^2+x+5\right)\frac{\pi}{2}\right) - \cos\left(\left(x^2+x+5\right)\pi\right)\right)$ is equal to _____. **[JEE(Main)April,2023]**

यदि $S=\left\{x \in \mathbb{R}: \sin^{-1}\left(\frac{x+1}{\sqrt{x^2+2x+2}}\right)-\sin^{-1}\left(\frac{x}{\sqrt{x^2+1}}\right)=\frac{\pi}{4}\right\}$, है, तो

$\sum_{x \in s}\left(\sin\left(\left(x^2+x+5\right)\frac{\pi}{2}\right)-\cos\left(\left(x^2+x+5\right)\pi\right)\right)$ बराबर है _____. **[JEE(Main)April,2023]**

Ans. D

Sol. $\sin^{-1}\left(\frac{x+1}{\sqrt{x^2+2x+2}}\right)-\sin^{-1}\left(\frac{x}{\sqrt{x^2+1}}\right)=\frac{\pi}{4}$

$\tan^{-1}(x+1)-\tan^{-1}x=\frac{\pi}{4} \Rightarrow \tan^{-1}\left(\frac{(x+1)-x}{1+(x+1)x}\right)=\frac{\pi}{4}$

$\tan^{-1}\left(\frac{1}{1+x^2+x}\right)=\frac{\pi}{4} \Rightarrow \frac{1}{1+x^2+x}=1$

$x^2+x+1=1 \Rightarrow x^2+x=0$

$\Rightarrow x=0,-1 \Rightarrow S=\{0,-1\}$

Now, $\sum_{x \in S}\left(\sin\left(\left(x^2+x+5\right)\right)\frac{\pi}{2}\right)-\cos\left(\left(x^2+x+5\right)\pi\right)$

$=\left(\sin\frac{5\pi}{2}-\cos 5\pi\right)+\left(\sin\frac{5\pi}{2}-\cos 5\pi\right)=(1+1)+(1+1)=4$

15. For $x \in (-1, 1]$, the number of solutions of the equation $\sin^{-1}x = 2\tan^{-1}x$ is equal to _____.
[JEE(Main)April,2023]

$x \in (-1, 1]$ के लिए, समीकरण $\sin^{-1}x = 2\tan^{-1}x$ के हलों की संख्या है _____ **[JEE(Main)April,2023]**

Ans. B

Sol. $\sin(\sin^{-1}x) = \sin(2\tan^{-1}x)$

$x=\frac{2x}{1+x^2}$

$x = 0, \pm 1$

$\Rightarrow x = 0$ and 1 are possible

16. Considering only the principal values of inverse trigonometric functions, the number of positive real values of x satisfying $\tan^{-1}(x)+\tan^{-1}(2x)=\frac{\pi}{4}$ is : **[JEE(Main)Jan.2024]**

(A) 1 (B) 2 (C) 0 (D) more than 2

प्रतिलोम त्रिकोणमितीय फलनों के मुख्य मान लेने पर, $\tan^{-1}(x)+\tan^{-1}(2x)=\frac{\pi}{4}$ को संतुष्ट करने वाले x के धनात्मक वास्तविक मानों की संख्या है : **[JEE(Main)Jan.2024]**

(A) 1 (B) 2 (C) 0 (D) 2 से अधिक

Ans. A

Sol. $\tan^{-1} x + \tan^{-1} 2x = \frac{\pi}{4}; x > 0$

$\Rightarrow \tan^{-1} 2x = \frac{\pi}{4} - \tan^{-1} x$

Taking tan both sides

$\Rightarrow 2x = \frac{1-x}{1+x}$

$\Rightarrow 2x^2 + 3x - 1 = 0$

$x = \frac{-3 \pm \sqrt{9+8}}{8} = \frac{-3 \pm \sqrt{17}}{8}$

Only possible $x = \frac{-3+\sqrt{17}}{8}$

17. Let $x = \frac{m}{n}$ (m, n are co-prime natural numbers) be a solution of the equation $\cos\left(2\sin^{-1} x\right) = \frac{1}{9}$ and let α, β ($\alpha > \beta$) be the roots of the equation $mx^2 - nx - m + n = 0$. Then the point (α,β) lies on the line : **[JEE(Main)Jan.2024]**

माना समीकरण $\cos\left(2\sin^{-1} x\right) = \frac{1}{9}$ का एक हल $x = \frac{m}{n}$ (m, n असहभाज्य घनपूर्णांक हैं) है और माना समीकरण $mx^2 - nx - m + n = 0$ के मूल α, β ($\alpha > \beta$) हैं। तो बिन्दु (α,β) किस रेखा पर है : **[JEE(Main)Jan.2024]**

(A) $5x + 8y = 9$ (B) $5x - 8y = -9$ (C) $3x + 2y = 2$ (D) $3x - 2y = -2$

Ans. A

Sol. $\sin^{-1} x = \theta \in I^{st}$ quadrant

$\cos 2\theta = \frac{1}{9}$

$\Rightarrow 1 - 2\sin^2\theta = \frac{1}{9}$

$\Rightarrow \sin^2\theta = \frac{4}{9}$

$\Rightarrow x^2 = \frac{4}{9} \Rightarrow x = \frac{2}{3} = \frac{m}{n}$

So $mx^2 - nx - m + n = 0$

$\Rightarrow 2x^2 - 3x + 1 = 0$

roots are $1, \frac{1}{2}$

$\alpha = 1, \beta = \frac{1}{2}$

So $(\alpha, \beta) = \left(1, \frac{1}{2}\right)$ lies or $5x + 8y = 9$

18. If the domain of the function $f(x) = \cos^{-1}\left(\frac{2-|x|}{4}\right) + \{\log_e(3-x)\}^{-1}$ is $[-\alpha, \beta) - \{\gamma\}$, then $\alpha + \beta + \gamma$ is equal to:

[JEE(Main)Jan.2024]

यदि फलन $f(x) = \cos^{-1}\left(\frac{2-|x|}{4}\right) + \{\log_e(3-x)\}^{-1}$ का प्रांत $[-\alpha, \beta) - \{\gamma\}$ है, तो $\alpha + \beta + \gamma$ बराबर है :

[JEE(Main)Jan.2024]

(A) 9 (B) 11 (C) 12 (D) 8

Ans. B

Sol. $-1 \le \left|\frac{2-|x|}{4}\right| \le 1$

$\Rightarrow \left|\frac{2-|x|}{4}\right| \le 1$

$-4 \le 2 - |x| \le 4$

$-6 \le -|x| \le 2$

$-2 \le |x| \le 6$

$|x| \le 6$

$\Rightarrow x \in [-6, 6]$ ...(1)

Now, $3 - x \neq 1$

And $x \neq 2$ ...(2)

and $3 - x > 0$

$x < 3$ ...(3)

From (1), (2) and (3)

$\Rightarrow x \in [-6, 3) - \{2\}$

$\alpha = 6$

$\beta = 3$

$\gamma = 2$

$\alpha + \beta + \gamma = 11$

19. For $\alpha, \beta, \gamma \neq 0$, if $\sin^{-1}\alpha + \sin^{-1}\beta + \sin^{-1}\gamma = \pi$ and $(\alpha + \beta + \gamma)(\alpha - \gamma + \beta) = 3\alpha\beta$, then γ equals :

[JEE(Main)Jan.2024]

$\alpha, \beta, \gamma \neq 0$, के लिए, यदि $\sin^{-1}\alpha + \sin^{-1}\beta + \sin^{-1}\gamma = \pi$ है तथा $(\alpha + \beta + \gamma)(\alpha - \gamma + \beta) = 3\alpha\beta$ है, तो γ बराबर है :

[JEE(Main)Jan.2024]

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\sqrt{3}$ (D) $\frac{\sqrt{3}-1}{2\sqrt{2}}$

Ans. B

Sol. Let $\sin^{-1}\alpha = A$, $\sin^{-1}\beta = B$ and $\sin^{-1}\gamma = C$

$\therefore A + B + C = \pi$

Also, $(\alpha + \beta + \gamma)(\alpha - \gamma + \beta) = 3\alpha\beta$

$\Rightarrow (\alpha+\beta)^2 - \gamma^2 = 3\alpha\beta$

$\Rightarrow \alpha^2 + \beta^2 - \gamma^2 = \alpha\beta$

$\Rightarrow \frac{\alpha^2 + \beta^2 - \gamma^2}{2\alpha\beta} = \frac{1}{2}$

$\Rightarrow \cos C = \frac{1}{2} \Rightarrow C = 60°$

$\therefore \sin C = \frac{\sqrt{3}}{2} = \gamma$

20. If $a = \sin^{-1}(\sin(5))$ and $b = \cos^{-1}(\cos(5))$, then $a^2 + b^2$ is equal to : **[JEE(Main)Jan.2024]**

यदि $a = \sin^{-1}(\sin(5))$ तथा $b = \cos^{-1}(\cos(5))$, तो $a^2 + b^2$ बराबर है : **[JEE(Main)Jan.2024]**

(A) $4\pi^2 + 25$ (B) $4\pi^2 - 20\pi + 50$ (C) $8\pi^2 - 40\pi + 50$ (D) 25

Ans. C

Sol. $a = \sin^{-1}(\sin 5) = 5 - 2\pi$

and $b = \cos^{-1}(\cos 5) = 2\pi - 5$

$\therefore a^2 + b^2 = (5 - 2\pi)^2 + (2\pi - 5)^2$

$= 8\pi^2 - 40\pi + 50$

21. If the domain of the function $\sin^{-1}\left(\frac{3x-22}{2x-19}\right) + \log_e\left(\frac{3x^2-8x+5}{x^2-3x-10}\right)$ is $(\alpha, \beta]$, then $3\alpha + 10\beta$ is equal to :

[JEE(Main)April,2024]

यदि फलन $\sin^{-1}\left(\frac{3x-22}{2x-19}\right) + \log_e\left(\frac{3x^2-8x+5}{x^2-3x-10}\right)$ का प्राँत $(\alpha, \beta]$ है, तो $3\alpha + 10\beta$ बराबर है :

[JEE(Main)April,2024]

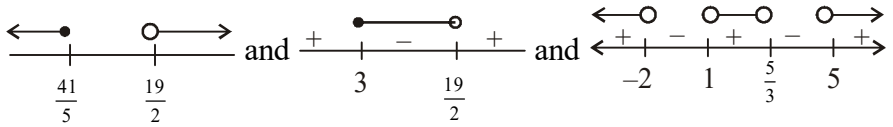
(A) 97 (B) 98 (C) 100 (D) 95

Ans. A

Sol. $-1 \le \frac{3x-22}{2x-19} \le 1$ and $\frac{3x^2-8x+5}{x^2-3x-10} > 0$

$\frac{3x-22}{2x-19} \ge -1$ and $\frac{3x-22}{2x-19} \le 1$ and $\frac{(x-1)(3x-5)}{(x-5)(x+2)} > 0$

$\Rightarrow \frac{5x-41}{2x-19} \ge 0$ and $\frac{x-3}{2x-19} \le 0$ and $\frac{(x-1)(3x-5)}{(x-5)(x+5)} > 0$



$x \in \left(5, \frac{41}{5}\right]$ Hence $3\alpha + 10\beta = 15 + 82 = 97$

22. Given that the inverse trigonometric function assumes principal values only. Let x, y be any two real numbers in [–1,1] such that $\cos^{-1}x - \sin^{-1}y = \alpha$, $\frac{-\pi}{2} \le \alpha \le \pi$. Then the minimum value of $x^2 + y^2 + 2xy \sin\alpha$ is :

[JEE(Main)April,2024]

दिया है कि प्रतिलोम त्रिकोणमति फलन केवल मुख्य मान मानता है। माना [–1,1] में दो वास्तविक संख्याऐं x तथा y इस प्रकार है कि $\cos^{-1}x - \sin^{-1}y = \alpha$, $\frac{-\pi}{2} \le \alpha \le \pi$ तो $x^2 + y^2 + 2xy \sin\alpha$ का न्यूनतम मान है : **[JEE(Main)April,2024]**

(A) –1 (B) $\frac{1}{2}$ (C) $\frac{-1}{2}$ (D) 0

Ans. D

Sol. $\cos^{-1}x - \left(\frac{\pi}{2} - \cos^{-1}y\right) = \alpha$

$\cos^{-1}x + \cos^{-1}y = \frac{\pi}{2} + \alpha$

$\alpha \in \left[-\frac{\pi}{2}, \pi\right], \frac{\pi}{2} + \alpha \in \left[0, \frac{3\pi}{2}\right]$

$\cos^{-1}\left(xy - \sqrt{1-x^2}\sqrt{1-y^2}\right) = \frac{\pi}{2} + \alpha$

$xy - \sqrt{1-x^2}\sqrt{1-y^2} = -\sin\alpha$

$(xy + \sin\alpha)^2 = (1 - x^2)(1 - y^2)$

$x^2y^2 + 2xy \sin a + \sin^2 a = 1 - x^2 - y^2 + x^2y^2$

$x^2 + y^2 + 2xy \sin\alpha = 1 - \sin^2\alpha$

$x^2 + y^2 + 2xy \sin\alpha = \cos^2\alpha$

Minimum value of $\cos^2\alpha = 0$

At $\alpha = \frac{\pi}{2}$

23. If the domain of the function $f(x) = \sin^{-1}\left(\frac{x-1}{2x+3}\right)$ is $\mathbf{R} - (\alpha, \beta)$, then $12\alpha\beta$ is equal to :

[JEE(Main)April,2024]

यदि फलन $f(x) = \sin^{-1}\left(\frac{x-1}{2x+3}\right)$ का प्रांत $\mathbf{R} - (\alpha, \beta)$ है, तो $12\alpha\beta$ बराबर है : **[JEE(Main)April,2024]**

(A) 24 (B) 36 (C) 40 (D) 32

Ans. D

Sol. $-1 \le \frac{x-1}{2x+3} \le 1$

$\frac{x-1}{2x+3} \ge -1$

$\frac{3x+2}{2x+3} \ge 0$

$x \in \left(-\infty, \frac{-3}{2}\right) \cup \left[-\frac{2}{3}, \infty\right)$ .....(i)

$\frac{x-1}{2x+3} \le 1$

$\frac{-x-4}{2x+3} \le 0$

$\frac{x+4}{2x+3} \ge 0$

$x \in (-\infty, -4] \cup \left(-\frac{3}{2}, \infty\right)$ .....(ii)

(i) $\cap$ (ii)

$x \in (-\infty, -4] \cup \left[-\frac{2}{3}, \infty\right)$

$\therefore x \in R - \left(-4, -\frac{2}{3}\right)$

$12\alpha\beta = 12 \times (-4) \times \left(-\frac{2}{3}\right) = 32$

24. For $n \in N$, if $\cot^{-1} 3 + \cot^{-1} 4 + \cot^{-1} 5 + \cot^{-1} n = \frac{\pi}{4}$, then n is equal to _____.

[JEE(Main)April,2024]

$n \in N$ के लिए, यदि $\cot^{-1} 3 + \cot^{-1} 4 + \cot^{-1} 5 + \cot^{-1} n = \frac{\pi}{4}$, तो n बराबर है _____

[JEE(Main)April,2024]

Ans. 47

Sol. $\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\frac{1}{4} + \tan^{-1}\frac{1}{5} = \frac{\pi}{4} - \tan^{-1}\frac{1}{n}$

$\tan^{-1}\left(\frac{\frac{1}{3}+\frac{1}{4}}{1-\frac{1}{12}}\right) + \tan^{-1}\frac{1}{5} = \frac{\pi}{4} - \tan^{-1}\frac{1}{n}$

$\tan^{-1}\frac{7}{11} + \tan^{-1}\frac{1}{5} = \frac{\pi}{4} - \tan^{-1}\frac{1}{n}$

$\tan^{-1}\left(\frac{\frac{7}{11}+\frac{1}{5}}{1-\frac{7}{55}}\right) = \frac{\pi}{4} - \tan^{-1}\frac{1}{n}$

$\frac{46}{48} = \tan\left(\frac{\pi}{4} - \tan^{-1}\frac{1}{n}\right)$

$\frac{46}{48} = \frac{n-1}{n+1}$

$n = 47$

25. Let the inverse trigonometric functions take principal values. The number of real solutions of the equation $2\sin^{-1} x + 3\cos^{-1} x = \frac{2\pi}{5}$ is _____. **[JEE(Main)April,2024]**

माना प्रतिलोम त्रिकोणमितीय फलन मुख्य मान लेते हैं। तो समीकरण $2\sin^{-1}x+3\cos^{-1}x=\frac{2\pi}{5}$ के वास्तविक हलों की संख्या है _____.

[JEE(Main)April,2024]

Ans. 0

Sol. $2\sin^{-1}x+2\cos^{-1}x+\cos^{-1}x=\frac{2\pi}{5}$

$$2\left(\frac{\pi}{2}\right)+\cos^{-1}x=\frac{2\pi}{5}$$

$$\cos^{-1}x=\frac{2\pi}{5}-\pi$$

$$\cos^{-1}x=\frac{-3\pi}{5}$$

But $\cos^{-1}x \in [0, \pi]$

$\therefore$ No real value of x

No solution